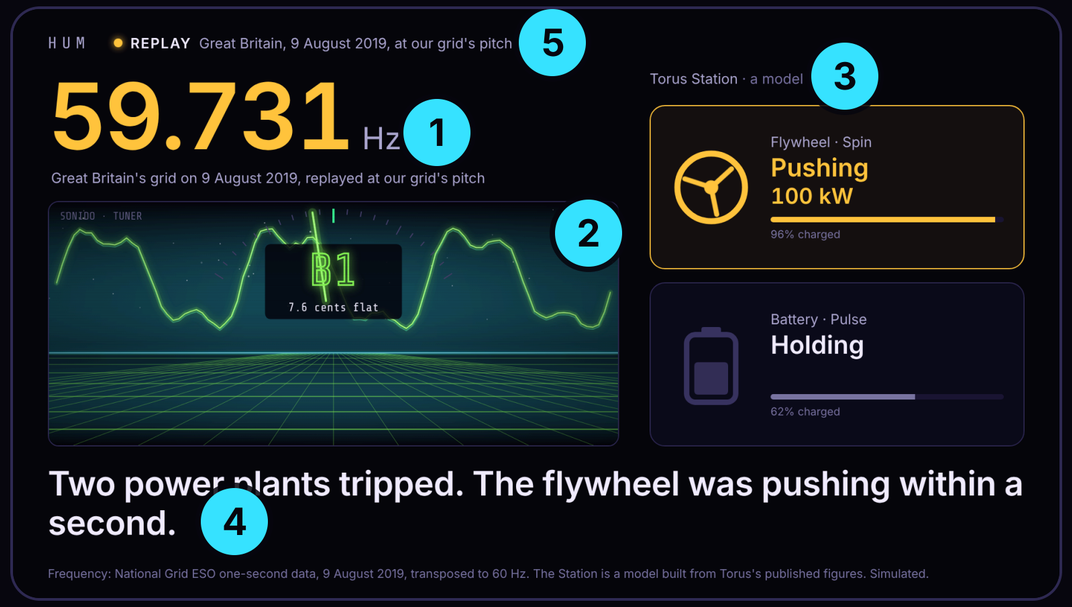
Morgan Espitia, Salt Lake City · Hard Tech Week 2026 · Torus bounty · Nucleus × Startup State open call

HUM hears the power grid through a guitar, and a model of a Torus Station answers it.

1. **The number** is the grid's frequency in hertz. It holds near 60 while supply meets demand and falls when the grid runs short of power.
2. **The tuner** shows the same hum as a musical note. At 60 Hz it reads B, in tune.
3. **The two cards** are the model Station. Its flywheel answers in the first seconds, and its battery carries the building for hours.
4. **The sentence** says what is happening and what the Station did about it.
5. **The tag** says where the reading comes from. LIVE is a room's wiring, TEST HUM is a synthesized hum in its place, and REPLAY is Britain's real grid failure of 9 August 2019.



What comes next: the live screen in four situations (a steady grid, tomorrow's rush hour, a real grid failure and a blackout), then what the compiler proves about the software that decided, then the plan.

What you will see

ampactor.dev/hum

Data: grid frequency, NESO open data at 1 s, 9 Aug 2019, moved to 60 Hz · rush hour, EIA-930 hourly demand, PacifiCorp East, 19–26 Oct 2026 · Station: a simulated model of Torus's published figures

How it works

The power grid's frequency shows whether it has enough power. A guitar can hear it.

The grid The whole Western grid, from Alberta to Baja, runs at one frequency: 60 hertz. When demand outruns supply, the frequency falls.

The ear Every power adapter hums at that frequency. A guitar pickup resting on one hears it, and my audio library, sonido, measures it to a thousandth of a hertz.

The Station A model of a Torus Station: a flywheel for the first seconds (100 kW for six minutes) and a battery for the hours (300 kW for four hours). Simulated.

The software Decides what the Station does. It is written in Mentl, a language I built: its rule, never reach the internet, is in the code, and the compiler checks it through every function the decision calls.

Station figures: Torus's published specifications, modeled and simulated. Frequency: a room's wiring when the demo runs live; the hosted recording uses a synthesized 60 Hz hum.

AI now helps write the software that runs machines. C and C++, the languages most of it is written in, can't check what that software will never do.

Batteries, flywheels and flight controls run on code that people test and review, and then trust.

The rule is part of the code, and the compiler checks it through every call.

You write the rule into the code: this never touches the internet, this never writes a file, this never allocates memory.

The compiler checks the rule through every function the code calls. **Break it, and the program won't build.**

Seven months and 2,600 commits since March. The compiler is written in Mentl and compiles itself. Open source: github.com/ampactor-labs/mentl

The compiler's answer about the controller in this demo

```
$ mentl why examples/grid/station.mn decide Network
decide never uses Network: nothing in decide, and
nothing it calls, performs an operation of Network

$ mentl why examples/grid/station.mn decide Alloc
decide never uses Alloc: nothing in decide, and
nothing it calls, performs an operation of Alloc
```


Counts: the git history of github.com/ampactor-labs/mentl. The answers: pitch/why.txt, captured from the compiler on 28 Oct 2026.

Everything you just saw was built since Monday at 4 PM, on two tools I built before it.

Before Mentl, the language: 2,610 commits since March 15. sonido, the audio library: 461 commits since January 25.

This week All of HUM: the ear, the controller, the director, the page and this show. In sonido, 10 commits, among them the grid ear and the pedal. In Mentl, 27.

Check it Every commit, with its time: github.com/ampactor-labs/hum, printed by `./ledger.sh`

Counts: each repository's git history since Monday 26 Oct 2026, 4:00 PM MDT, printed in pitch/ledger.txt.

Who needs it in Utah

TORUS · SALT LAKE CITY

Builds flywheel-and-battery Stations that steady the grid. The software inside each Station is where a proof would matter first. Torus is judging this demo.

HILL AFB · 309TH SOFTWARE ENGINEERING GROUP

Maintains flight software for the F-16, the F-35 and the missile fleet, where every claim about the software needs evidence.

The sites I'll ask to host the pilot: Torus, Rocky Mountain Power's Wattsmart Battery program, and Utah State University's ASPIRE center.
First hires: a controls engineer and a compiler engineer, in Salt Lake City.

What $10,000 unlocks

$10,000 puts this controller beside a real battery in Utah within ninety days.

Day 30 Form the Utah company. Run the controller on a 3 kW inverter and 10 kWh of batteries on my bench.

Day 60 With a site's written consent, run it read-only beside a real battery for 14 days, logging its decisions and sending no commands.

Day 90 A pilot agreement that lets it send commands, and an SBIR proposal with Nucleus Grow.

$7,000 for the bench. **$3,000** for the company, a lawyer for the pilot agreement, a year of insurance and the site's read-only kit. Every line is priced in the budget.

Three asks. A Utah operator who will host it. An introduction to Hill AFB's 309th. A co-founder who knows power electronics.

Thank you, Torus and Nucleus

Replay this demo on your phone.



`ampactor.dev/hum`

Morgan Espitia · sonofagoose@pm.me

The demo: github.com/ampactor-labs/hum
The language: github.com/ampactor-labs/mentl

Three asks: a Utah operator who will host the pilot, an introduction to Hill AFB's 309th, and a co-founder who knows power electronics.

What Mentl code looks like

Five operators draw the shape of a program. The compiler checks each rule along that shape.

`let l = saw |> rig.dark(300.0) |> soft_clip`	`|>`	in sequence: the sawtooth, the filter, the clipper
`let (echoed, roomed) = dry <| (rig.echo, rig.room)`	`<|`	one sample to the echo and to the room; shared, so the two cannot race
`let l = ((echoed) >< (dry)) |> mix(0.35)`	`><`	two signals side by side, then mixed
`{ prev => a * x + (1.0 - a) * prev } <~ delay(1)`	`<~`	feedback: this sample from the last one; the memory is allocated once
`handler rack = spectral_flux_distort ~> limiter_at(0.6)`	`~>`	the handler: what the sound goes through, named once

The controller's rule is on its decision: `with !Alloc + !Network + !Filesystem + !WASI + !Trap`. The compiler checks it through every function the decision calls, and the copy that read a price online did not build.

Lines from lessons 2, 3 and 4 of the Mentl course and from examples/grid/station.mn, as they check clean on 29 Oct 2026. The language: github.com/ampactor-labs/mentl